

7. Feedback loops are essential to homeostasis. **Image 7.1** shows a generalised feedback loop and **Image 7.2** is a photograph of a person suffering from a condition caused by excessive alcohol consumption.

Image 7.1

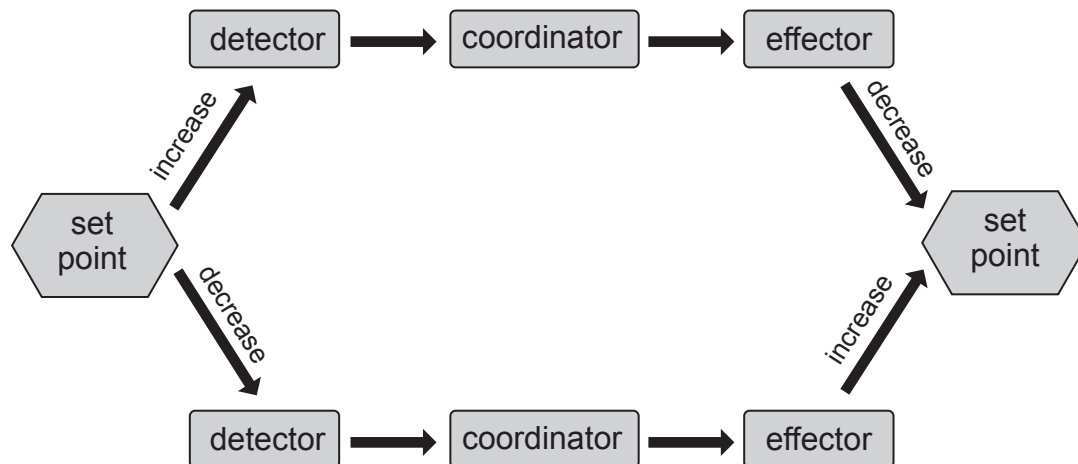


Image 7.2



Oedema: excess fluid retention.

This condition can occur if the solute concentration of blood is too low, meaning that the water potential is not negative enough. This means that water cannot be drawn back into the blood from the surrounding tissue fluid.

Explain the functions of each of the **three** components of a generalised feedback loop shown in **Image 7.1**.

With reference to the generalised feedback loop shown in **Image 7.1**, describe the role of ADH in osmoregulation.

Long-term alcohol use has been shown to cause thickening of the basement membrane of glomeruli. Explain how this change could cause the condition shown in **Image 7.2**. (Details of tissue fluid formation are not required.) [9 QER]



Examiner
only

Area for writing answers, consisting of horizontal dotted lines.

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